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EDUCATION

Cornell University **–** Cornell David A. Duffield College of Engineering **Ithaca, NY**
B.S. in Mechanical Engineering. *Expected 2030*

International Baccalaureate (Frederic Mistral – Tècnic Eulàlia) **Barcelona, Spain**
Relevant coursework: *Physics HL*, *Math AA HL* & *Business Management SL* *Sept. 2024 – May 2026*
Grades: 39/45 (3.92/4.00 GPA)

Harvard Undergraduate Ventures-TECH Summer Program **(HUVTSP '25) —** 6-week entrepreneurship and emerging tech program with 50+ international students.

PROJECTS

Aerodynamic Comparison of the 2022 and 2026 Formula 1 Car Models *Sept. 2025 – Nov. 2025*

- Mentored by Georgia Tech Ph.D. provided by Lumiere Education.
- Applied computational (CFD) methods to model airflow dynamics around both models.
- Compared the models from 2022 and 2026 to see if the new car is more suitable to Wake Effects.
- Collected data on drag, cross-sectional area and downforce using digital instrumentation and proposed a new way to evaluate how energetic is a car's wake ($Wake\ Rating = C_d \cdot A \cdot p_{max}$).
- Concluded that the 2026 model produces a less energetic and more controlled wake, consistent with the FIA's effort to promote closer racing by reducing the car's wake effects.
- Under the defined Wake Rating, the 2026 model produced a rating of 2,174.11, compared with 3,414.86 for the 2022-25 regulation reference model. This represents a 36% reduction in Wake Rating under the study's assumptions.

F1 Aerodynamics: How does the angle of attack of the front wing affect "$C_d \cdot A$" *Mar. 2025 – Nov. 2025*

- Designed and conducted an experiment to quantify how Formula 1 front-wing AoA affects $C_d \cdot A$.
- Built a scale LEGO 2024 F1 car with an adjustable front wing and developed a controlled wind-tunnel testing setup.
- Used Tracker video analysis and Newton's Second Law to calculate acceleration, relative airspeed, $C_d \cdot A$, and experimental uncertainty.
- Identified a linear relationship between front-wing AoA and $C_d \cdot A$ (slope ≈ 0.000004 / degree, $R^2 = 0.952$), and proposed improvements to better align the conclusion with theoretical assumptions.

EXPERIENCE

DocuBridge **Remote**
Technical Intern *Jul 2025 – Aug 2025*

- Collaborated in a 5-person team to develop an AI-powered web tool for analyzing Excel-based financial models (P&L, Balance Sheet, Cash Flow…)
- Delivered 4+ project milestones, including backend logic, UI prototype and API documentation, resulting in implementation on DocuBridge's platform.
- Applied agile development practices and presented final project to the company's CEO.
- Engineered API integration and backend data-processing logic.

SKILLS & INTERESTS

Tools: SimScale Computational Fluid Dynamics (CFD), Microsoft Excel, Tracker.
Currently building: SolidWorks (self-directed CAD, aero assembly design), Python (CS1112 at Cornell) & MATLAB.
Languages: Catalan (Native), Spanish (Native), Italian (Native), English (C2) & French (B2)
Interests: Football, skiing, Spanish reggaeton music, Formula 1 and investing.